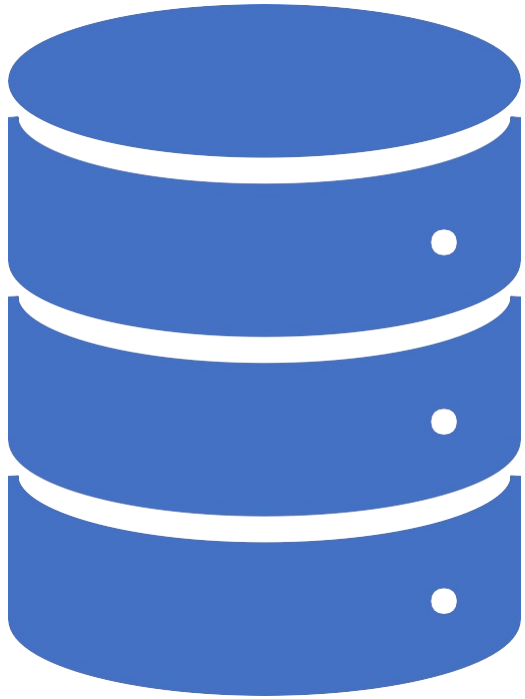
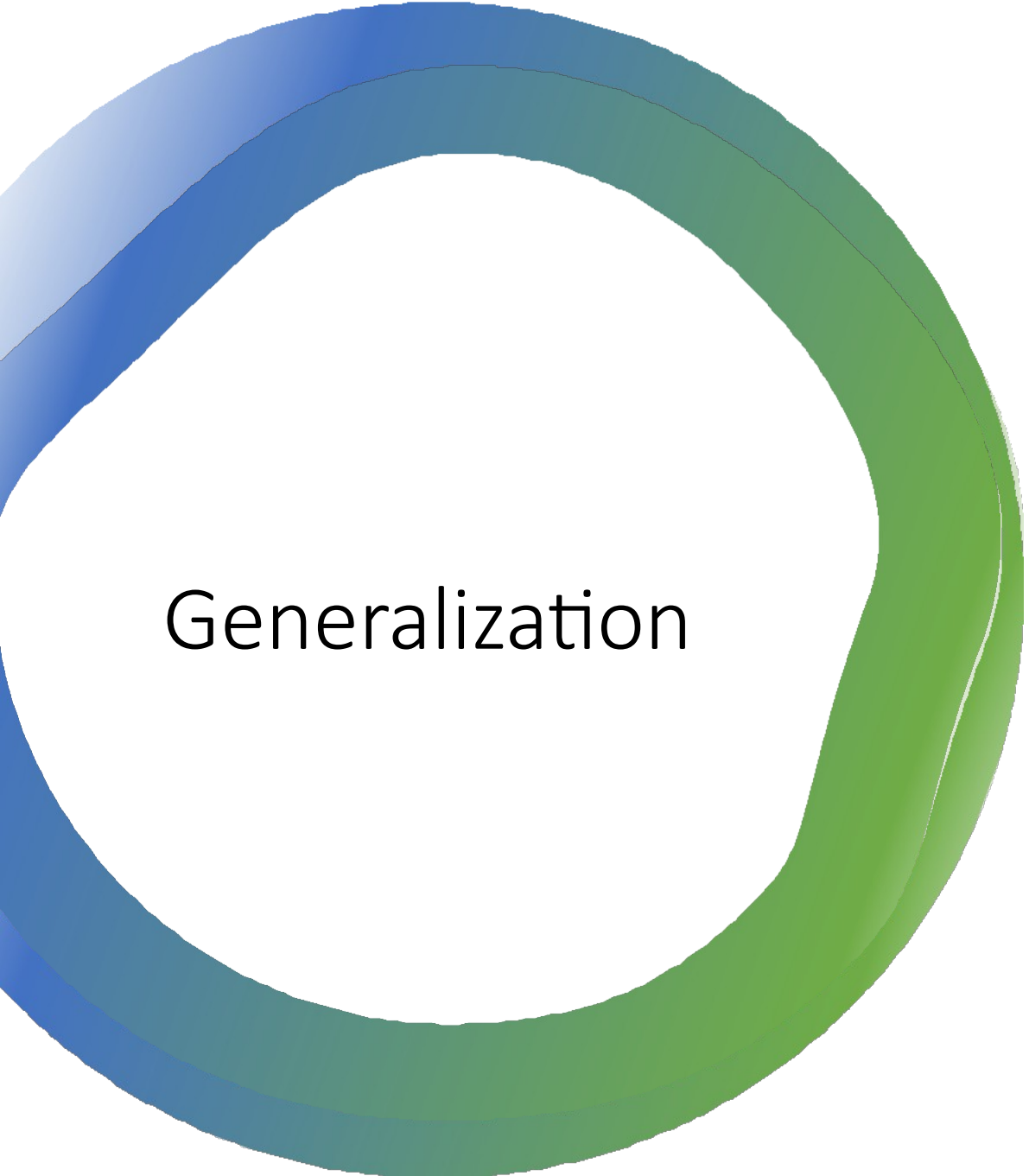


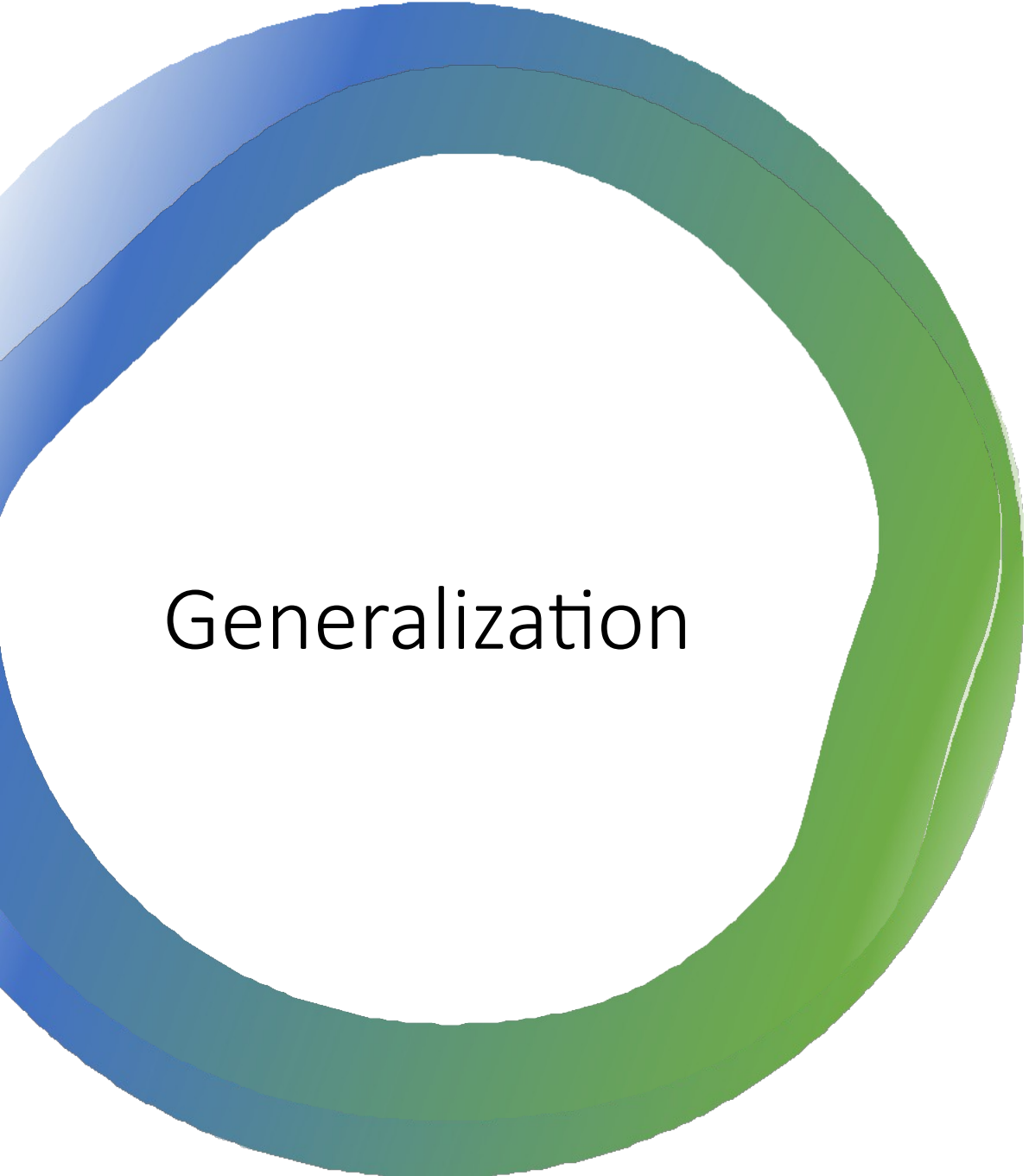


Database System
[*Generalization,*
Specialization and
Aggregation]



Generalization

- **Generalization** is like a bottom-up approach in which two or more entities of lower level combine to form a higher level entity if they have some attributes in common.
- In generalization, an entity of a higher level can also combine with the entities of the lower level to form a further higher level entity.

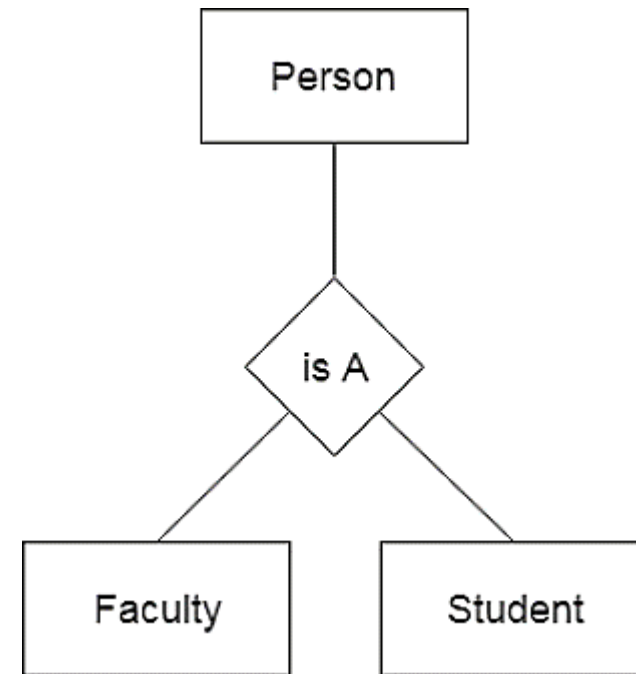


Generalization

- Generalization is more like subclass and superclass system, but the only difference is the approach. Generalization uses the bottom-up approach.
- In generalization, entities are combined to form a more generalized entity, i.e., subclasses are combined to make a superclass.

Generalization Example

- Faculty and Student entities can be generalized and create a higher level entity Person.



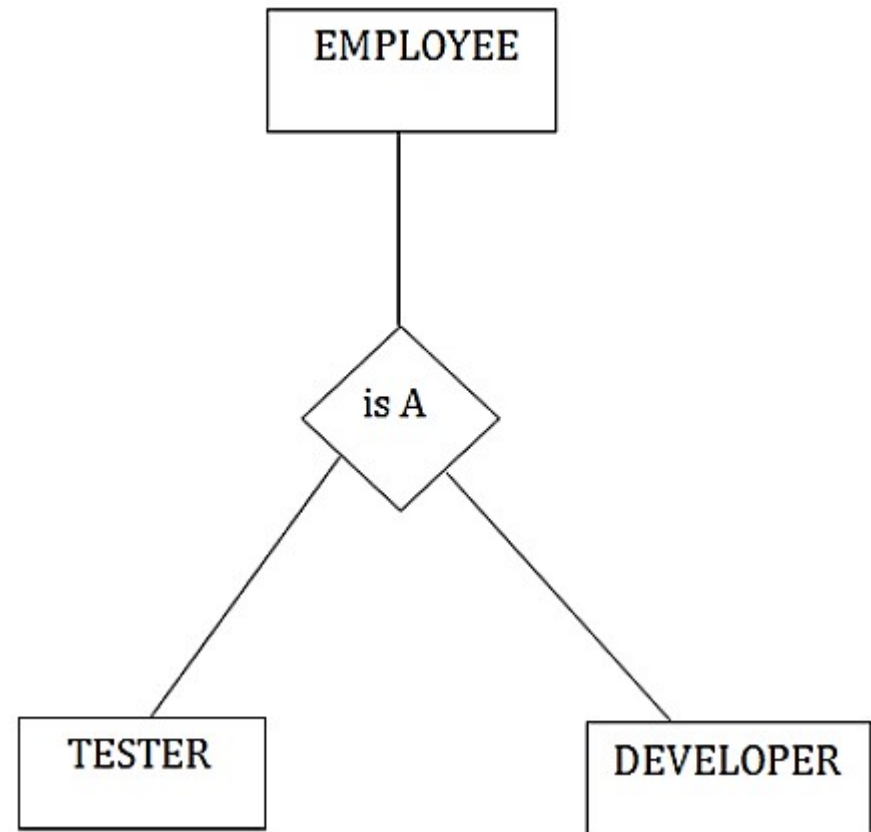


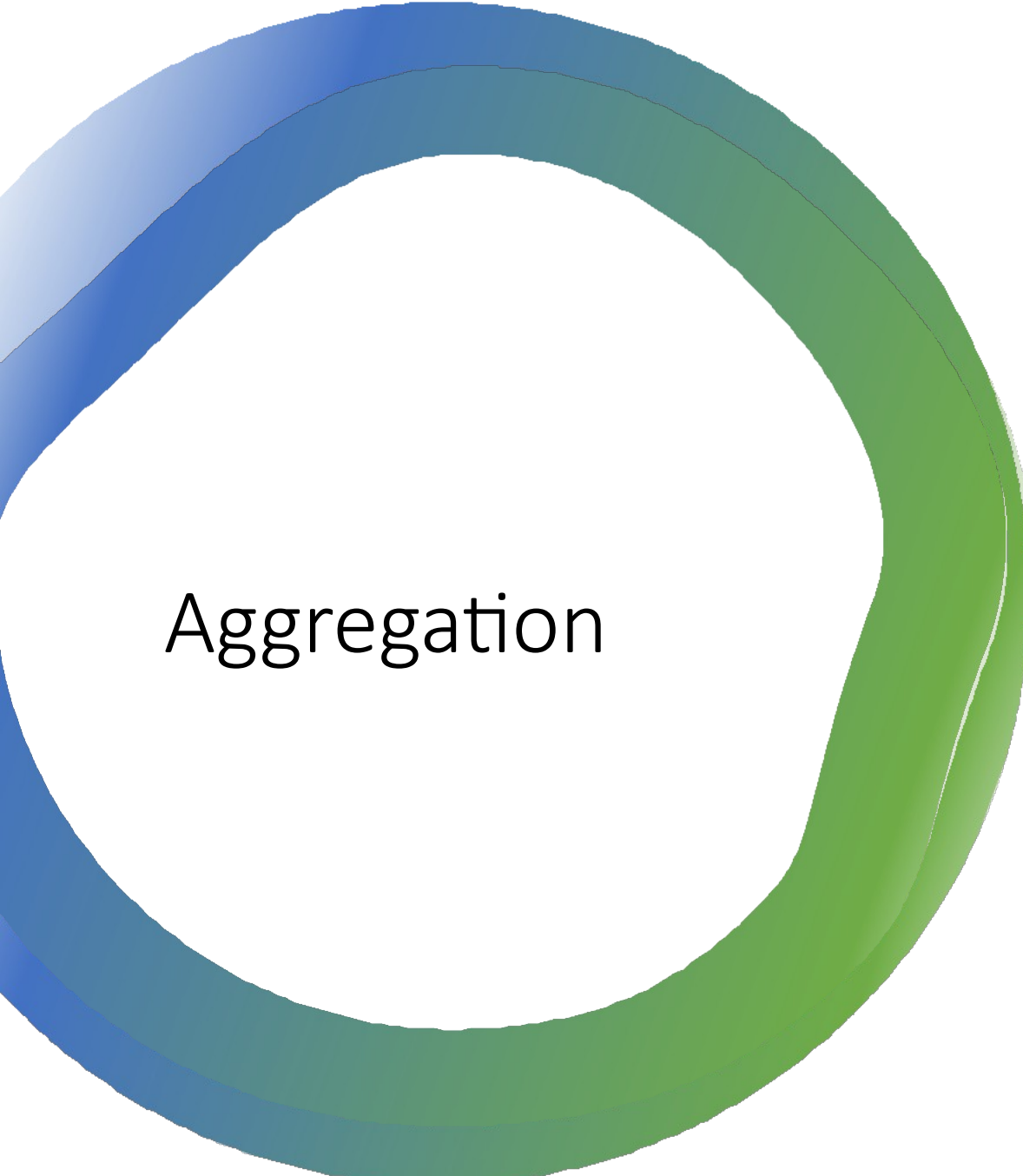
Specialization

- **Specialization** is a top-down approach, and it is opposite to Generalization. In specialization, one higher level entity can be broken down into two lower level entities.
- Specialization is used to identify the subset of an entity set that shares some distinguishing characteristics.
- Normally, the superclass is defined first, the subclass and its related attributes are defined next, and relationship set are then added.

Specialization Example

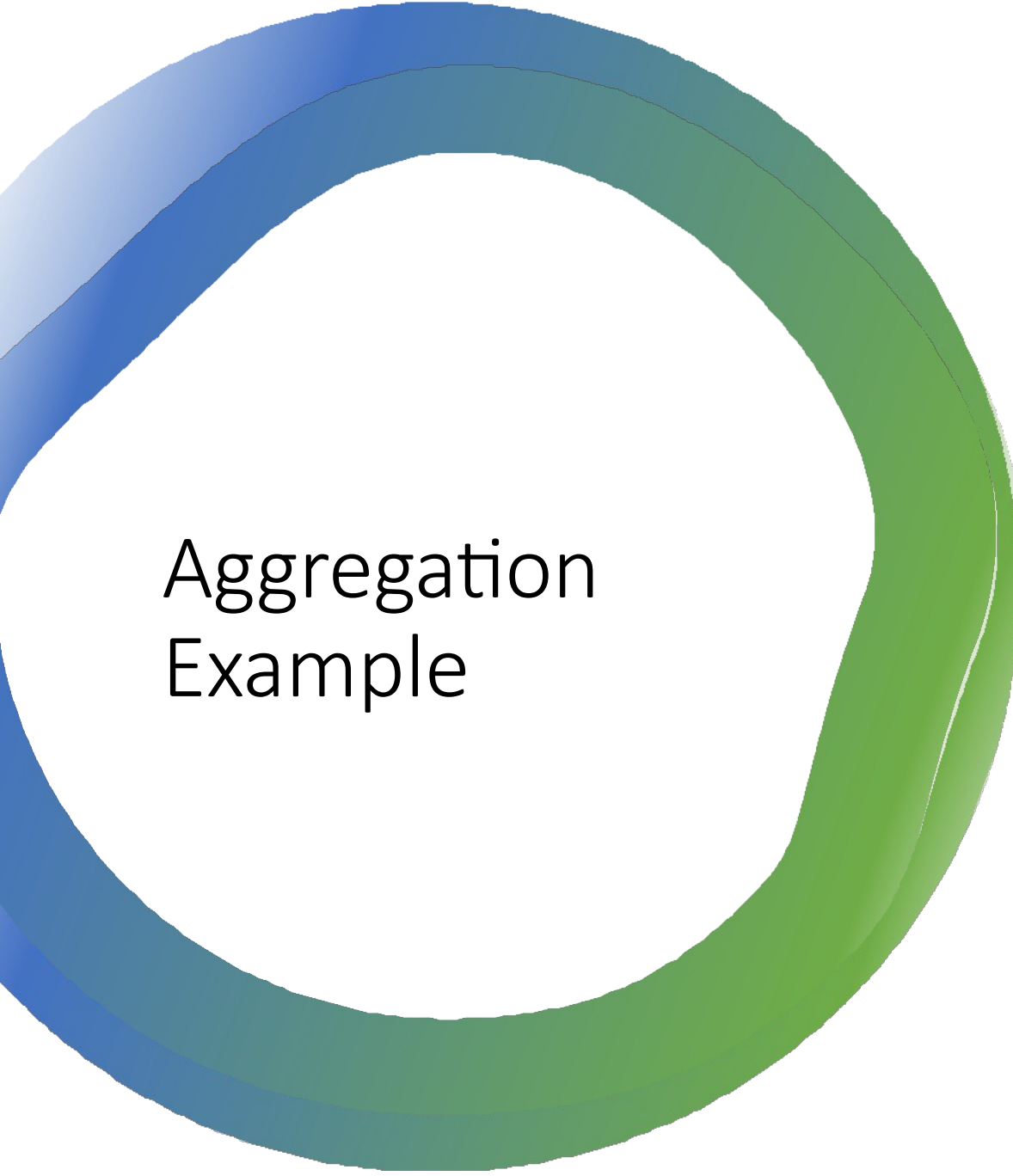
- In an Employee management system, EMPLOYEE entity can be specialized as TESTER or DEVELOPER based on what role they play in the company.





Aggregation

- In **aggregation**, the relation between two entities is treated as a single entity.
- In aggregation, relationship with its corresponding entities is aggregated into a higher level entity.



Aggregation Example

- **Center** entity offers the **Course** entity act as a single entity in the relationship which is in a relationship with another entity **Visitor**.
- In the real world, if a **Visitor** visits a coaching center then he will never enquiry about the **Course** only or just about the **Center** instead he will ask the enquiry about both.

Aggregation Example

